FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
KING ABDULAZIZ UNIVERSITY

FCIT
KAU

كلية الحاسبات وتقنية المعلومات
جامعة الملك عبدالعزيز

King Abdulaziz University

Faculty of Computing & Information Technology

Final Lab Exam (11/12/2025)
Course: CPCS 204 (Data Structures 1)
Semester: Fall 2026, Time: **9:00 pm – 10:30 pm**

Grade: _____ / 15 Points

Name: ________________________ **ID:** _________________ **Section:** ______

Note: Read the following instructions very carefully!

1. From **Blackboard**, download **CPCS_204_FinalLabExam_Fall_2025.zip** stored in **Quizzes/Tests** option.
2. Save the .zip file on **Desktop** or at your desired location.
3. Unzip this file by right clicking on this folder. Then delete previous .zip file.
4. Open **NetBeans** and select the option **File➡Open Project…** from the main menu.
5. Open all classes of your lab exam (4 in total) and **write your ID, Name** etc on top of every class and start writing the code in the classes given below.
6. Make sure you keep on saving your files as you work so you don't lose it.
7. When you finish your exam, go to Blackboard and upload your solution after zipping the folder which contains your solution.
8. **You as a student are responsible for uploading the right solution folder on Blackboard. No invigilator is responsible to do so.**
9. You can add or delete any parameter in any class according to your requirements.

Question 1:	3.5

In choice 1 in the main class:

1) Write a code to **reverse ONLY the first half of the given Stack S1** by using a queue, for this purpose use only ‘pop’, ‘push’, ‘enqueue’ and ‘dequeue’ methods.
2) Show the stack S1 contents before and after your code.

```
|-------------------------------------------------------------|
|---------------     Queue - Stack (Menu)    ----------------|
|-------------------------------------------------------------|
|   1. Reverse the first half of Stack S1                     |
|   2. Delete negative Elements of Stack S2                   |
|   3. Divide the values of new Stack                         |
|   4. Count Multiples                                        |
|   5. Quit                                                   |
|-----------------------------------------------------------|

> Please enter your choice: 1
Stack S1 Contents BEFORE reversing are:
10, 9, 8, 7, 6, 5, 4, 3, 2, 1,

Elements of Stack S1 AFTER reversing are:
6, 7, 8, 9, 10, 5, 4, 3, 2, 1,
```

Question 2:	3.5

In choice 2 in the main class:

1) Write a code to **delete the negative elements of the given Stack S2** by using another Stack, for this purpose use only ‘pop’ and ‘push’ methods for this purpose.
2) Show the stack S2 contents before and after your code

```
|-------------------------------------------------------------|
|---------------     Queue - Stack (Menu)    ----------------|
|-------------------------------------------------------------|
|   1. Reverse the first half of Stack S1                     |
|   2. Delete negative Elements of Stack S2                   |
|   3. Divide the values of new Stack                         |
|   4. Count Multiples                                        |
|   5. Quit                                                   |
|-----------------------------------------------------------|

> Please enter your choice: 2
Elements of Stack S2 BEFORE deleting the negative elements are:
-15, 5, -12, 4, -9, 3, -6, 2, -3, 1,

Elements of Stack S2 AFTER deleting the negatine elements are:
5, 4, 3, 2, 1,
```

Question 3:	4

In choice 3 in the main class:

1) Create a new Stack that filled with 25 values increased by 3 (3,6,9,12,15,18,……..etc) and then print its contents.
2) Implement the method **EvenToHalf()** in **StackLL class** :
   This method should **divide** the value of each stack node by 2 *only if the value is even* and then RETURN the number of changed nodes.
3) Apply the **EvenToHalf()** on this stack and show the results
4) Print the stack after calling the method to show the changes.

```
|------------------------------------------------------------|
|--------------     Queue - Stack (Menu)    ----------------|
|------------------------------------------------------------|
|   1. Reverse the first half of Stack S1                    |
|   2. Delete negative Elements of Stack S2                  |
|   3. Divide the values of new Stack                        |
|   4. Count Multiples                                       |
|   5. Quit                                                  |
|----------------------------------------------------------|

> Please enter your choice: 3

Stack Contents  :
75, 72, 69, 66, 63, 60, 57, 54, 51, 48, 45, 42, 39, 36, 33, 30, 27, 24, 21, 18, 15, 12, 9, 6, 3,
Stack Contents after the changes :
75, 36, 69, 33, 63, 30, 57, 27, 51, 24, 45, 21, 39, 18, 33, 15, 27, 12, 21, 9, 15, 6, 9, 3, 3,
The Number of even elements is: 12
|------------------------------------------------------------|
```

Question 4:	4

In choice 4 in the main class:

1) Create a new Queue that filled with 7 values that taken from the user and then print its contents.
2) Implement the method **countMultiples()** in **QueueLL class** :
   This method should **count** the values that are multiple of 4, multiple of 5, not multiple of both 4 and 5, and then it returns the results as an array
3) Apply the **countMultiples()** on this Queue and show the results

```
|--------------     Queue - Stack (Menu)    ----------------|
|------------------------------------------------------------|
|   1. Reverse the first half of Stack S1                    |
|   2. Delete negative Elements of Stack S2                  |
|   3. Divide the values of new Stack                        |
|   4. Count Multiples                                       |
|   5. Quit                                                  |
|----------------------------------------------------------|

> Please enter your choice: 4

> Please enter 7 integers values : 10 11 12 13 14 15 16
Elements of the Queue are:
10, 11, 12, 13, 14, 15, 16,

Number of valuses that are multiple of 4 :2
Number of valuses that are multiple of 5 :2
Number of valuses that are not multiple of 4 and 5 : 3
```

Best Wishes